em lyon business school

CENTRALE LYON



we are makers

DIPLÔME CONFÉRANT GRADE DE LICENCE CONTRÔLÉ PAR L'ÉTAT

Bachelor of Science in Data Science for Responsible Business



Isabelle HUAULT
Executive President & Dean
***emlyon** business school*

*In a complex world made of socio-environmental issues, **emlyon** business school is on a mission to provide life-long training and support, to informed people, who can readily transform organizations for a fairer society, with more solidarity and respect for the planet.*

In that respect, the School's strategy was constructed based on three main aspects: social and environmental commitment, network internationalization and academic excellence.

The Bachelor of Science "Data Science for Responsible Business" fully embraces this new orientation and addresses the needs for a responsible training hybridizing high standard knowledge.

*By renewing the historical synergies between **emlyon** business school and **Centrale Lyon**, this program is now a track where disciplines no longer walk side by side, but where they merge together for mutual-fertilization purposes.*

As a true hybrid track, the Bachelor of Science aims at developing cross-functional skills in keeping with responsibility, and the technical and scientific competences of students, based on the interconnections of learnings and the way they are put into perspective. This program combines the best technical and managerial teachings for a training up to the level of the issues of responsible data.



Jean SAVINIEN
Academic Director
***emlyon** business school*

AI impacts us in many profound ways and invites us to rethink the relations and traditional boundaries which exist between science, business, and human and social questions.

We are convinced that responsible commitment is necessary when working in AI and that it requires the creation of bridges between engineering and business units.

Our program targets high-potential students and delivers a hybrid advanced training in data science and management to fill a gap in organizations, and to answer their need for responsible and enlightened AI experts.



Pascal RAY
*Director **Centrale Lyon***

*Institutions of higher education must be key players leading the transition regarding environmental, energetic, digital and social matters. **Centrale Lyon** decided to build its strategy around these challenges. The creation of the Transition Lab, a complete ecosystem on the matters symbolizes this strategy, whose recognition is shown by our position in national and international rankings.*

Digital and data sciences are another strength of our School: Yohann DE CASTRO, scientific referee of the Bachelor and teacher-researcher in mathematics has also been a member of the Institut Universitaire de France since October 1, 2022.

Industry 4.0, predictive medicine, aeronautics, cybersecurity, but also banking, insurance or tourism: many sectors need skills in data management and analysis, artificial intelligence and machine learning.

*Today, it is fundamental to reconcile digital sciences and social responsibility; the question of uses and impact must be inherent in the development of these new technologies. This is why, thanks to the cross-disciplinary views of **emlyon** and **Centrale Lyon**, we are offering a transdisciplinary educational approach. The hybrid course we have designed not only provides the technical skills and managerial know-how that are essential, but also transmits the tools and attitudes that will enable young graduates to act as visionary, committed, and responsible actors.*



Zied BOUYAHYA
Academic Co-Director
Centrale Lyon

Using data science helps businesses communicate clearly with their stakeholders and swiftly adjust to a changing regulatory landscape. This ensures they stay competitive and build long-term trust. By integrating data science technologies and recent advances in AI, companies not only enhance their operations but also foster a culture of accountability and integrity, which is of paramount importance for responsible business practices.

By joining this program, you will acquire the technical knowledge and analytical skills necessary to excel in this dynamic field. From predictive modeling to natural language processing, big data and deep learning, we cover the full spectrum of data science disciplines.

We are excited to have you on board and look forward to supporting your development and success in this academic journey and beyond.



Key points

emlyon business school

Top 1% of business schools worldwide

accredited
AACSB, EQUIS, AMBA

#10
In Europe
European Business School Ranking 2025 Financial Times

139
articles per year in international scientific papers

9,865
students

4 campuses
Lyon, Shanghai, Paris and Mumbai

52,000
graduates in 130 countries

Centrale Lyon

+3,200
students

330
teacher-researchers and teachers

12,4 M€
research agreement

6
laboratories
joint research units with CNRS

+900
articles per year in international scientific journals

2025 figures

Bachelor of Science *Data Science for Responsible Business*

DIPLÔME CONFÉRANT GRADE DE LICENCE CONTRÔLÉ PAR L'ÉTAT

4-year post baccalaureate program leading to a Bachelor's degree

Key points

The **Bachelor of Science Data Science for Responsible Business** is a 4-year Bachelor's Degree of excellence in data science and management. For those looking to pursue with higher education, this program can be a passageway to some of the best Master's Degrees. The Bachelor of Science targets **high-potential students** and delivers a **hybrid advanced training in data science and management** to fill a gap in organizations and answer their need for responsible and enlightened AI experts.

The pedagogy of the program is applied and action oriented (individual and group projects, business games, flipped classrooms...). Students will learn the best professional practices of AI and get a real know-how of data science through concrete business projects and responsible AI missions within companies, organizations, and research labs.

The Bachelor of Science in Data Science for Responsible Business program allows for the validation of the Level 6 professional certification "Sciences et ingénierie, Sciences du management - Sciences des données pour une entreprise responsable," issued by **emlyon** business school and registered with the RNCP under number 40435, registered on 09.01.2024.

This program is eligible for VAE (APEL: accreditation of prior experiential learning).

For what type of students?

For high-school or Baccalaureate students and graduates with a specialty in Mathematics or Science

Program highlights

- Dual skills in business and engeneering
- Integrated pedagogy
- Concrete vision of professionnal and human impacts of AI and data science
- Management of concrete business projects
- Strong internationalization
- Numerous internships
- Supportive alumni community

Program structure

Year 1	Mathematics & Computer Science Fundamentals	3-month internship in NPO
	Fundamentals of Responsible Business	
Year 2	Data Science | Technologies of AI	3-month internship in tech
	Business Practice & Corporate Social Responsibility	
Year 3	Semester Abroad Advanced Language Models | Modern Trends in AI	3-month track internship
	Digital Business | Corporate Law	
Year 4	Research Missions | Tech Seminars	6-9 month internships BSc Thesis
	Innovation | Human & Machine Cognition	

Discover the program in videos

Research Track

Data Scientist Track

Entrepreneurship Track

It is possible to validate this diploma by blocks of competencies.
Assessment methods may include case studies, presentations, mock exercices, tests, etc.

Program tracks

Data scientist

Students work in partnership with a company on a full-scale data science project during a year, join the company during their summer internship, and deliver their solution to the company.

Research

Students work in partnership with a research lab on a research project tutored by a researcher.

They join the lab during their summer internship. Partner research labs include the **“Institut Camille Jordan”** (mathematics) and **“Laboratoire d’InfoRmatique en Image et Systèmes d’Information”** (computer science) from the University of Lyon.

Entrepreneurship

Students are trained and supported by the incubators of **emlyon** business school and **Centrale Lyon** to start their own businesses. They work full time developping their company during their summer intership.

The graduation rate for this program is not available, as the program was launched in 2023.



Build
your future career

Each one of you is unique, with your own personality, desires, commitments, development potential and experience. This is why we offer you personalized support to advise you in your studies, as well as in your choice of internships and career paths.

1. Comprehensive support

Our Career Services department provides individual support to help you build your personal and professional project in order to ensure your employability.

2. Professionalizing experience

With at least 18 months spent in companies, you will build and fine-tune your professional project, in France and abroad, in order to successfully join the job market.

Employment data and statistics are not yet available for this program (first intake in 2023).

3. Each year be inspired by leaders of the industry

Each year, honorary patrons lend their expertise and commitment to the program, inspiring students with real-world perspectives. Through their testimonies, they share their vision of the challenges and opportunities shaping the industry.

Honorary patron of the class of 2024/2025:



Vincent NGUYEN
EMEA System Architect at NVIDIA

Artificial intelligence represents an industrial revolution of immense magnitude, triggering a transformative era that, while marked by shortterm uncertainties, ultimately unlocks tremendous global opportunities. As the landscape of work evolves, some specific roles will inevitably disappear, yet they will be replaced by entirely new job categories that remain beyond our current imagination.

Honorary patron of the class of 2024/2025:



Téma BENHALIMA-BOUVILLE
PhD Global Innovation Director at ENGIE

This program represents a major step toward professionalizing a scientific field that directly serves industrial business cases. Though students may come from very different backgrounds, they ultimately converge toward exploring the endless possibilities of data and the positive impact it can have on innovation and energy transition when used responsibly and with discernment, for a neutral and circular economy.

Honorary patron of the class of 2026/2027:



Chouaieb NEMRI
Generative AI Blackbelt Specialist at Google

Generative AI is redefining our world at a staggering pace, urging us to rethink the traditional boundaries between science, business, and society. As an AI practitioner, I see firsthand the critical need for leaders who can bridge deep technical expertise with ethical responsibility. The hybrid approach of this Bachelor of Science between emlyon and Centrale Lyon perfectly addresses this gap. I challenge this new promotion to not only master the data, but to engineer the impossible for a sustainable and responsible future.



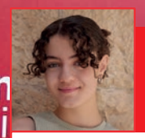
Sofia
3rd year student from Russia

This program gives me the opportunity to develop professionally in a multicultural society. I can meet people from all around the world with diverse cultures and backgrounds. It helps me to become open-minded to new ideas and perspectives. What I like about my degree, is the number of different opportunities you are given. All the skills you are acquiring can help you to bring almost any business project to the next level by adding the power of data.



Boudsalakham
2nd year student from Laos

Being double enrolled in an engineering school and a business school makes up a big part of the learning experience, but there is more to it. We also have established experts who come to teach us the same data science methods they employ in their day-to-day work. And because we primarily learn by creating personal projects, by the end of our second year we have built a strong portfolio by industry standards, and are ready to take on technical roles as part of our summer internships.



Selina
2nd year student from Israël

What makes this program distinctive is the intensity of its international and professional exposure. Studying in a multicultural environment, collaborating with students from diverse backgrounds, and following courses entirely in English have helped me develop a strong global mindset. The program is demanding, with internships required each year and opportunities such as an exchange semester further enriching the experience. This balance between academic rigor and continuous real-world experience makes the journey both challenging and motivating, while preparing us to confidently pursue international careers.



Arora
1st year student from India

*My degree allows me to experience two distinct campus environments at **Centrale Lyon** and **emlyon** Business School, **Centrale Lyon** offers a rigorous, technically grounded setting built on mathematics and data driven courses, while **emlyon** provides a modern business-oriented atmosphere focused on real-world skill courses. I truly enjoy having access to both campus's vibrant and diverse student associations.*

How can I join the program?

Apply online on

Admission

You are eligible to the Bachelor of Science in Data Science for Responsible Business if you are:

- A holder or future holder of the **French baccalaureate with a specialty in Mathematics or Science**: through the French procedure Parcoursup
- A holder or future holder of an **international diploma** accepted as equivalent to the **baccalaureate with a specialty in Mathematics or Science:** through our online admission test

This program is eligible for VAE (APEL: accreditation of prior experiential learning), find all the information on em-lyon.com

Tuition fees

For September 2027
Tuition fees: €15,050 per year
Student services fees: €450 per year
Total tuition fees: €15,500 per year

The stated annual tuition and fees apply to students entering the program in the 2027-2028 academic year and remain unchanged for the duration of their four-year program. This rate is subject to change for students entering in subsequent years. The comprehensive fee includes tuition, student services and enrollment fees, academic expenses, and lifelong memberships to both the **emlyon** business school and the **Centrale Lyon** Alumni networks.



You are temporarily or permanently disabled? Please contact our teams at handicap@em-lyon.com to study how you can integrate the program

You will find the welcome guide and all the information on how we welcome and support our students and participants with disabilities at em-lyon.com/en

For personalized information, please contact our advisors:

Admission Department
Tel: +33 4 12 05 86 90
Email: bscdata@em-lyon.com

emlyon campus

By joining the Bachelor of Science in Data Science for Responsible Business, you will take courses on both campuses: **emlyon** business school and **Centrale Lyon**.

A campus in the center of Lyon

The **emlyon** campus is located in Lyon's 7th arrondissement, at the center of a dynamic ecosystem of companies, universities, and socio-economic stakeholders. This central location fosters exchanges, partnerships, and professional opportunities throughout your program.

Spaces designed for collaboration

Over 30,000 m², the campus offers spaces dedicated to student life and learning: auditorium, library, makers' lab, entrepreneurship spaces, cafés, and collaborative work areas. These spaces stimulate creativity, innovation, and experiential learning.

Designed for a responsible future

Designed around principles of sustainability, the **emlyon** campus reflects the school's commitment to a responsible future. Its spaces encourage sustainable innovation and international openness, making the campus a true hub of inspiration and diversity.





Centrale Lyon

Located in Écully, on the outskirts of Lyon, the **Centrale Lyon** campus offers a remarkable environment that combines academic excellence and quality of life. Nestled at the heart of a green setting, the campus provides an ideal environment to fully immerse yourself in your studies and your student life.

A green campus

Covering more than 17 hectares of green spaces, the **Centrale Lyon** campus offers a natural environment conducive to focus, reflection, and personal development. Sports facilities, collaborative spaces, and relaxation areas allow students to study in a calm and environmentally friendly setting. A true asset for a balanced academic and personal experience.

At the heart of scientific research & innovation

Centrale Lyon is a campus deeply rooted in scientific and technological research. Students work alongside researchers and engineers within high-level laboratories, recognized nationally and internationally for their expertise in fields such as artificial intelligence, complex systems, and applied mathematics.

A strategic location

Just a few minutes from downtown Lyon, students benefit from the dynamism of a major European metropolis while studying in a calm, safe and welcoming environment.



Watch our video on the synergy between the two campuses for the BSc in Data Science students

emlyon business school campuses

LYON
144 avenue Jean Jaurès
69007 Lyon - France
em-lyon.com

SHANGHAI
Asia Europe Business School
155 Tan Jia Tang Road
Minhang District
Shanghai 201199 - People's Republic of China
www.em-lyon.com.cn

PARIS
15 boulevard Diderot
75012 Paris – France
em-lyon.com

MUMBAI
St Xavier College
5 Mahapalika Marg
Mumbai 400-001 - Maharashtra - India
xebs.edu.in

Centrale Lyon campus

36 avenue Guy de Collongue
69134 Ecully cedex - France
www.ec-lyon.fr



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